System Level Tests - LightSpeed 3.X

1 Module Overview

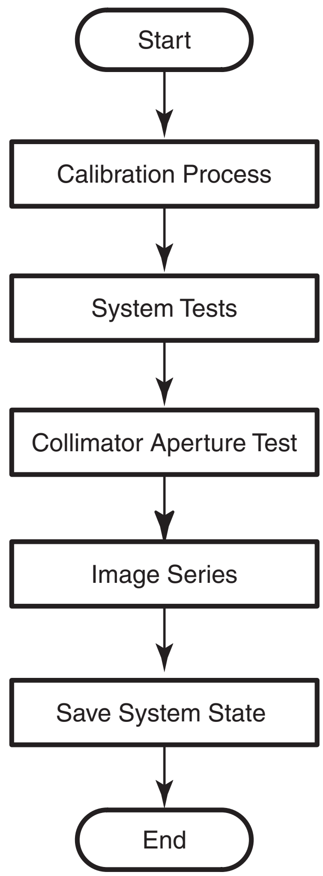
The System Level Tests for the LightSpeed 3.X include the following sections:

- Introduction
- Calibration Process
- System Tests
- Collimator Aperture Test
- Image Series
- Save System State

2 Introduction

This document contains descriptions of the procedures used to test system operation after installation, alignment and calibration, but prior to customer turnover and acceptance. The Image Series sections instruct you to scan system phantoms, record the means and standard deviations, and evaluate the image quality of the resulting images.

Figure 1: System Test Overview



3 Calibration Process

3.1 Reference Procedures

Do NOT perform these procedures now. They are placed here as Reference for later procedures. You will be directed to follow these instructions at several points throughout this document.

3.1.1 Scanning with Service Protocols

Locate the Manufacturing and Installation protocols under Infant area 20.

NOTE: **Manufacturing and Service share this Protocol list. Different product option offerings also use this list. Carefully follow the scan section instructions, and verify you acquired the images with the correct technique before filling out the data sheet. Otherwise you may troubleshoot an image problem that only exists because you used the wrong technique.**

1.) Select the NEW PATIENT icon.
2.) Enter a Patient ID (e.g., **`getest`**).
3.) Click on the box labeled PEDIATRICS.
4.) Select a service protocol from the list, to display the corresponding view edit screen.
   Optional Method: Enter the protocol number into the `Protocol Number Field` on the Exam Rx Screen.

3.1.2 Center Phantom

1.) Select SCANNER UTILITIES.
2.) Select CENTER PHANTOM.
3.) Follow the on-screen procedures.
4.) The phantom center spec is ±1mm.
5.) Select QUIT, when the phantom is within spec.

3.2 Prepare the QA Phantom

NOTE: **The QA phantom is shipped water filled.**

1.) Locate the multi-language sticker packet in the QA phantom shipping box.
2.) Attach the sticker with the customer's language to the face of the phantom hanger bracket.

3.3 Calibration Process Introduction

If your system has a factory supplied state MOD, you used it to load the system calibration files during the Restore System State.

3.4 Fast Cal

1.) Select 

to warm up the tube.

2.) Select FAST CALIBRATION from the Daily Prep menu.

NOTE: **Use the default Fast Cal selections determined by the system configuration. (The system defaults to all four kV stations, but you can choose kV stations to calibrate during reconfig.)**

3.) Run the selected air calibrations.

4.) When the calibration process completes, click on QUIT.

3.5 Table/Gantry Alignment Procedure

3.5.1 Conditions

This procedure may require the removal of the table bottom, side and rail covers.

This procedure requires that mechanical final alignment was completed and passed.

3.5.2 Procedure

1.) If this is a Non-Fixed Site, check the Van level. Correct, if necessary.

2.) Check Table level. Correct, if necessary.

3.) Drive the table to its highest elevation (phantom holder removed).

4.) Drive the table cradle in and out five times to seat the rollers.

5.) Turn on the alignment lights. Check the scan window for proper installation.

6.) Advance the end of the cradle (black dot on cradle) to the internal light.

7.) Complete Section 3.1.1, "Scanning with Service Protocols" now.

8.) Select the service protocol, PERPENDICULAR ALIGNMENT. Protocol should appear as shown in Table 1. Add two scouts to this procedure. 120kv 40 ma tube at zero, 120kv 40 ma tube at ninety.

Table 1: Perpendicular Alignment Scan Parameters

Scan Type	Direction	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Algorithm
Axial	Head First	120	40	Large	5mm/1i	2sec	S0	S0	Bone
Axial	Head First	120	40	Large	5mm/1i	2sec	I1000	I1000	Bone

Set Internal Landmark. The red boxes will disappear from the screen.

9.) Click on CONFIRM, then press the **START SCAN** button when lighted, to perform the scan.

10.) Use over/under multi-image display to view both images at the same time:

a.) Click on IMAGE WORKS

b.) Highlight the scans just performed and click on VIEWER.

c.) Click on FORMAT, and select the "over/under" view (two boxes aligned vertically).

11.) Use the upper image (cradle front) to verify that table is centered:

a.) Click on the "grid" icon.

b.) Click on MEASURE, and select the straight (angled) line.

c.) Using the mouse, position one end of the line on the vertical axis.

d.) Position the other end at one outside edge of the cradle, forming a perfectly horizontal line (angle = 90). Read the distance displayed on screen.

e.) Repeat step d to measure the distance from the vertical axis to the other edge.

12.) Repeat step 11 for the lower image (cradle at 1000 mm).

13.) If the edges are off by greater than 1 mm (front to back), perform the following:

a.) Loosen the table anchors at locations #5, #6 and #7.

b.) Move the table half the measured distance.

c.) Return to Step 8. Repeat the procedure until the table edges line up (± 1mm). If needed a scout can be completed to check table horizontal presentation image.

Check the two scout scans

i.) Check the tube at zero scan for table gantry perpendicularity. Make table adjustments as needed

ii.) Check the tube at 90 scan for cradle level and visual irregularity with the white side molding. Molding strips should be parallel within +/- 3 mm. Cradle should be level within ± 3 mm.

d.) Tighten the anchors and torque to 74.6 N-m (55 ft.-lb.) for fixed sites only. (75 ft-lb.) for non-fixed systems.

e.) Reinstall all table parts removed to access the anchors, including covers, ground straps, elevation motor leads and the center support bar.

14.) Install all table and gantry covers.

15.) Use ERASE ALL to remove lines and grids from the images.

16.) Click on END EXAM to exit the program.

3.6 Auto CT N Number Adjustment

NOTE: If it has been more than 30 minutes since calibration scans, rerun FASTCAL (approx 15 min). N Number Adjustment needs the tube to be warm, or CT numbers will change.

Run the small and large phantom CT N number adjustment before you acquire the image series, or any time you doubt the N# values. The calibration procedure modifies the pcal (phantom cal) vector to assign water a CT number equal to zero. After a successful N number adjustment, any water phantom scanned against a small or large CAL has a CT number near zero.

1.) Place the QA Phantom in the holder.

2.) From the application desktop, select SCANNER UTILITIES.

3.) Select AUTO CT NUMBER ADJUSTMENT.

4.) Follow directions on the screen.

a.) When centering the phantom:

* Centering may be easier to perform from the gantry right side.

* Center on the water portion of the phantom (not over the internal block).

b.) The program will automatically make adjustments. If the program is unable to make adjustments, it will stop at the technique it is unable to adjust. Troubleshoot with manual CT # Adjust before continuing.

4 System Tests

4.1 Introduction

Use the system tests in the following sections to exercise all aspects of the system, to assure system integrity, before turnover to the customer. Although the means, standard deviation, and resolution specifications do not apply during system functional tests, treat any artifact or image anomaly as a failure.

If you encounter a failure during the system tests:

- Record any evidence of artifacts, such as rings, streaks, shading, cupping, noise, or center artifacts.
- Correct artifacts, system test, or image series failures when they occur. Any delay in repairs could increase the number of retests.

4.2 Tomographic Plane Indication

1.) Place the QA phantom on the phantom holder.
2.) Turn ON the internal alignment lights, and drive the phantom into the gantry opening, until the line on the phantom lines up with the internal laser lights.
3.) Verify that BOTH internal axial lasers line up along the line on the QA phantom. If not, check table/gantry, cradle, and/or laser alignment.
4.) Center the phantom in the scan plane with the calibration program. (See 3.1.2, on page 3, for details on the phantom centering procedure.)
5.) Select the service protocol, TOMO PLANE INDICATION. (See 3.1.1, on page 3, for details on scanning with service protocols.)

or

Manually select the scan parameters in Table 2.

Table 2: Tomographic Plane Indication Scan Parameters

Scan Type	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Algorithm	Interval
Helical	120	200	Small	1.25HQ	1.0sec	I3.0	S3.0	Bone detail	0.2

6.) Display the image series, and locate the scan plane indicator, the longest bar in the bar pattern on the right side of the phantom. The right side of the phantom corresponds to the side of the image labeled **L** on the display screen.
7.) On the HHS Data Sheet, record the scan location (shown on the image annotation) of the image with the darkest scan plane indicator (darkest long bar).
8.) If your system meets all the installation and alignment specifications, the image at scan location zero (S0.0) should contain the scan plane indicator. If scan location S1.0 or scan location I1.0 has the darkest bar, the system still meets the specification. The scan plane deviation should equal S0.0 $\pm$1.0mm. If necessary, adjust the internal alignment light position to meet the S0.0 $\pm$1.0mm requirement.
9.) Repeat the Tomographic Plane Indication test with the external alignment lights.
    a.) Use the external alignment light, and press the external landmark.
    b.) Verify the external light lines up along the black line on BOTH the left and right sides of the QA phantom.
    c.) The scan plane indication must fall within the S0.0 $\pm$1.0mm specification.
10.) Check the box on **Form 4879**.

4.3 System Scanning Test

Use the System Scanning Test to verify hardware functionality. Review images for visible artifacts, and review the message log for unacceptable errors.

1.) Place the QA phantom on the cradle.
   - Drive the table to an elevation of 100.
   - Align the line on the phantom with the internal laser lights.

   Never scan above 50mA without first placing a phantom in the field of view. Levels in excess of 50mA can cause temporary radiation damage to the detector that lasts several hours. If you acquire image series cals with a radiation damaged detector, the cals may cause artifacts in subsequent image series scans.

2.) Select the service protocol, SYSTEM SCAN/CUST QUAL RELI.

NOTE: **Stop the service protocol after the second helical series. It is not necessary to proceed past that point in the protocol**.

-or-

Manually select the scan parameters specified in Table 3.

Table 3: System Scanning Test Scan Parameters

Scan Type	kV	mA	SFOV	Thickness	Rotation Time	Start Loc.	End Loc.	Tilt/Table Speed
Scout	120	40	-	-	-	S200	I800	0°
Scout	120	40	-	-	-	S200	I800	90°
Axial (2 scans)	120	50	Large	10mm/1i	1.0sec	S0	S0	I30
Axial (2scans)	120	50	Large	10mm/1i	1.0sec	S0	S0	S30
Helical HQ	120	50	Large	5mm	1.0sec	S70	I75	15mm/rot
Helical HS	120	50	Large	5mm	0.8sec	S70	I75	22.5mm/rot

3.) Complete the scans.

5 Collimator Aperture Test

5.1 Small Aperture Test

1.) Clear all obstructions out of the gantry SFOV.

2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon, .

3.) Click on the UTILITIES icon. 

4.) Select INSTALL.

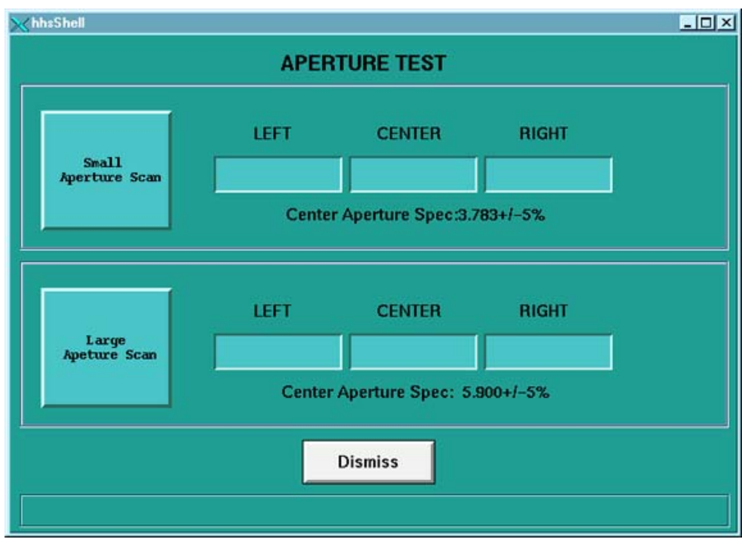
5.) Select COLLIMATOR APERTURE TEST.

6.) Select SMALL APERTURE SCAN to display a scan protocol window, and start gantry activity (refer to Figure 2).

7.) Press the **START SCAN** key when it flashes, to initiate the first scan.

8.) When the test completes, the screen updates to display the Small Aperture values.

Figure 2: Collimator Aperture Test Screen



5.2 Large Aperture Test

NOTE: The center channel is the only channel with a spec. The left and right channels may or may not pass. Small aperture center spec is 3.783 ± 5%. Large aperture center spec is 5.900 ± 5%.

1.) Select LARGE APERTURE to display a scan protocol window, and start gantry activity (Refer to Figure 2).

2.) Press the **START SCAN** key when it flashes, to initiate the first scan.

3.) When the test completes, the screen updates to display the large aperture values.

4.) Record the large and small aperture values on Form 4879.

6 Image Series

6.1 Scan Protocol

The person who acquires the image series has the responsibility to review the images, and verify they meet the specifications listed on data sheets. Responsibilities also include means and standard deviation measurements, and keeping a record of failures that occur during the image series.

Unless otherwise stated, use the following scan parameters during the image series acquisition:

- Scan FOV equal to display FOV (Field of View)
- 512x512 matrix size

NOTE: Consider *any* image series scan that does not meet specifications as failing.

For means and standard deviations, 90% of the slices must pass. Any failure on a particular technique requires at least ten additional slices to evaluate effectively.

Systems with metal-free cradles have a phantom holder with a perpendicular adjustment (Z-axis) knob on it. **Each time you change phantoms,** make sure you use a bubble level, and the Z-axis knob on the phantom holder, to level the phantom.

6.2 Data Recording: Means and Standard Deviation

Any failure on a particular technique requires at least a ten additional slices to evaluate effectively. For means and standard deviations, 90% of the slices must pass.

- Record means to two decimal places, and round to the nearest one-tenth, (one decimal place) when you compare the resulting values to the spec.
- Record standard deviations to two decimal places, then round off to one decimal place, to compare it to the spec.
- Average standard deviations: Use two decimal places to average the values, then round off to one place.

Before you record the means and standard deviations, check the image data sheets to determine whether to average the means and standard deviations, or record them slice by slice. Make sure you record all the required image data on the HHS data sheets.

6.3 Term Definitions & Screens

Xc - Mean CT number for the specified center coordinates of the phantom image.

AvXc - Average Mean CT number for the center of the phantom image: Average the mean CT value for all specified center coordinates of all slices in an exam.

Xo - Mean CT number for the outside of the phantom image: Average the mean CT value for all specified outside coordinates of one slice.

AvXo - Average outside mean CT number for the number of slices in an exam.

AvSDc - Average image noise about the center image coordinate (measured as the standard deviation) of all slices in an exam.

AvSDo - Average image noise (standard deviation) for the outside of a phantom: Average of all outside coordinates of all the slices in an exam.

Figure 3: Image Analysis Tool User Interface – Auto 1x Test Pull Down Menu

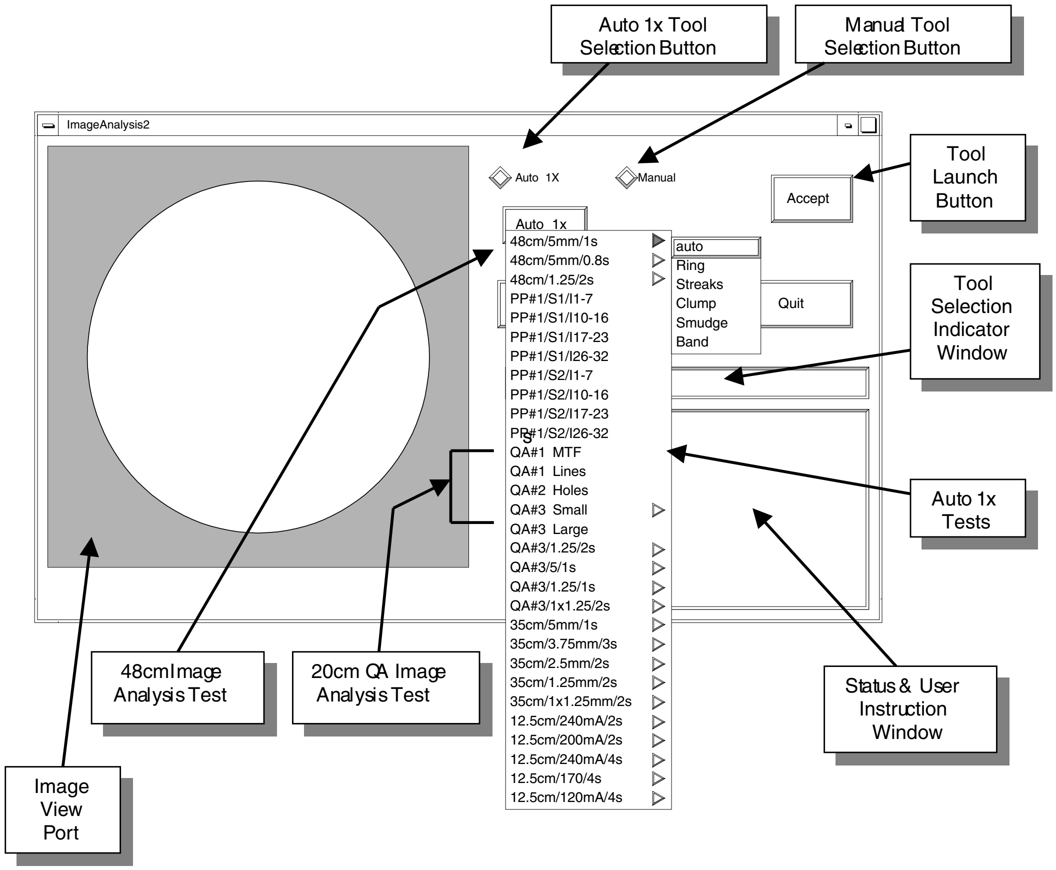
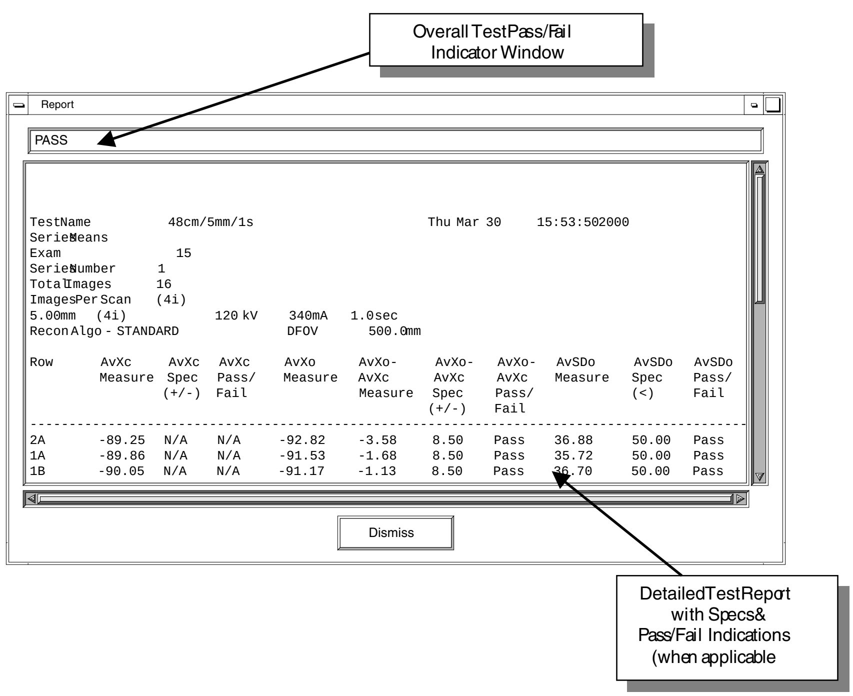


Figure 4: Analysis Tool User Interface – Test Results Report Window



6.4 48cm Phantom Image Series Image Performance Verification

6.4.1 Acquiring the 48cm Phantom Image Series

1.) Mount the Phantom Holder on the head-end of the table.
2.) Mount the 48cm Phantom on the Phantom Holder.
3.) Align, level, & center the 48cm Phantom.
   - Align phantom using the internal laser lights.
   - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
   - Center phantom using the CENTER PHANTOM utility in the left head SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.
4.) Set up the system to scan a four scan, 16 image, 48cm Phantom image series.

Manual Scan Protocol Setup

Refer to Table 4 and set-up an Axial scan with the parameters as shown.

Table 4: LightSpeed Plus 48cm Phantom Image Series Scan Parameters

Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
48 Lg Series 1	Axial Full 1.0 sec.	I7.50	S7.50	16	5.0 4i	0.00	S0.0	Large	120	340	4.0 sec.	50.0	Std

Auto Scan Protocol Setup

a.) On the Exam Rx desktop, select NEW PATIENT.
b.) Type the following entries in the two listed `Patient Information` fields:
   Patient ID: **`Service`**
   Name: **`48cm Phantom Image Series`**
c.) From the Protocol display, click on the infant box.
d.) On the infant display window, click on the area below the infant's right foot to display the `Miscellaneous` menu.
e.) Click on the 20:10 IMAGE SERIES 48 CM button.
f.) Select the first series (`Series Description` filed should display `48cm Lg Series 1`)

5.) Set internal Landmark.
6.) Acquire a four-scan, 16 image, 48 cm Phantom image series.

6.4.2 Brightness Uniformity and Noise

6.4.2.1 Image Performance Verification

1.) Select the 48cm Phantom image series exam acquired in the previous section.
   a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
   b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
   c.) Select the exam and series acquired in the previous section.
   d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.

2.) Build a 45 x 45 pixel reference ROI Box using the Image Works Viewer tools.

a.) Click on the grid button to place a grid on the first image.

b.) Click on the MEASURE button and select the box ROI icon.

c.) Adjust the size of the ROI box to be a square 45 mm x 45 mm (2025 mm2) box. Tolerance: 45 mm +/- 5 mm (1600 mm2 to 2500 mm2).

If required, magnify the image to adjust to proper dimensions. See Figure 5 for additional ROI size and placement information.

3.) Collect Mean and Standard deviation values for five reference ROI Box positions on the sixteen 48cm Phantom images.

If required, magnify the image to adjust to proper dimensions. See Figure 5 for additional ROI placement information.

a.) Position the reference ROI Box built in step 2 directly over the center of the image using the grid cross-hairs as a guide.

b.) Type **`prop a`** in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.

c.) Click on the MEASURE button and select the box ROI icon to display Box # 2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box # 1.

d.) Reposition Box # 2 to the left center portion on the first image (Box # 2 position in Figure 5).

e.) Type **`prop a`** in the Image Works Accelerator Line followed by <Return> to propagate Box # 2 ROI (size and position) on the remaining images.

f.) Repeat steps 3.c. through 3.e. for each of the remaining ROI box positions shown in Figure 5 (Box # 3 through 5).

g.) Record the Mean and Standard Deviation values of the 16 images in the series for each of the five box positions in Table 5.

Each image can only display text for the mean, standard deviation, and box area for three images at a time. To view the data for a particular box, select the box on the image and the system displays the data for the box number selected.

Figure 5: 48cm Phantom Brightness Uniformity & Noise Measurement – Building Placing Reference ROI Boxes

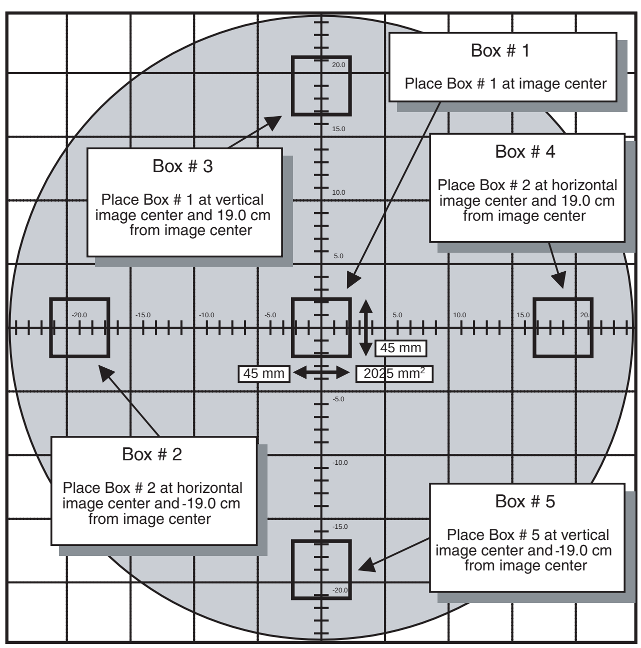


Table 5: 48cm Phantom CT# Brightness Uniformity & Noise Image Performance Worksheet

Image	Box 1 (Center)	Box 2 (Left Center)		Box 3 (Top Center)		Box 4 (Right Center)		Box 5 (Bottom Center)		Outside Boxes Avg	
	Means (Xc)	Means	Std dev	Means	Std dev	Means	Std dev	Means	Std dev	Means (AvXo)	Std dev (AvSDo)
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											
11											
12											
13											
14											
15											
16											

Box Size =1600 mm2 to 2500 mm2

45 mm (+/- 5 mm) x 45 mm (+/- 5 mm)

45 (+/- 4 pixels) x 45 (+/- 4 pixels)

Box Positions:Box 1 = 0 mm x 0 mm

Box 2 = 0 mm x -190 mm

Box 3 = 190 mm x 0 mm

Box 4 = 0 mm x 190 mm

Box 5 = -190 mm x 0 mm

Table 6: 48cm Phantom CT# Brightness Uniformity & Noise Row Performance Worksheet

Row	Images	Center Box Means (AvXc)	Outer Boxes Means Averages (AvXo)	Outer Boxes Means Averages minus Center Box Means (AvXo - AvXc)	Outer Boxes Standard Deviation Averages (AvSDo)	Comments
2A	1, 5, 9, 13					
1A	2, 6, 10, 14					
1B	3, 7, 11, 15					
2B	4, 8, 12, 16					
Specifications		n/a	n/a	± 8.5	< 70	

4.) Calculate the Brightness Uniformity and Noise values for each image in the series, record values in Table 6, and compare to specifications.

a.) Calculate and record the average means and average standard deviation values for the four outside Boxes (Boxes 2 through 5) for each of the images and record in Table 5.

b.) Calculate and record the center box (Box 1) average means values (AvXc) for each row in Table 6.

c.) Record the outside boxes (Box 2 through 5) average means values (AvXo) and average standard deviation values (AvSDo) for each row in Table 6.

d.) Calculate the Brightness Uniformity (AvXo - AvXc) value for each row and record in Table 6.

e.) Verify Brightness Uniformity (AvXo - AvXc) value and Noise value (AvSDo) for each row meets specifications listed in Table 6.

f.) Record the 48cm Phantom Brightness Uniformity (AvXo - AvXc) value and Noise value (AvSDo) for each row in the HHS Record Tables.

6.4.2.2 Failure Recovery

Specifications

Each Row (2A, 1A, 1B, and 2B) of the series *must* pass 48cm Brightness Uniformity and Noise (for the first series scan parameters) specifications:

- AvXo - AvXc: < +/- 8.5
- AvSDo: < 70.0

Recommended Recovery

A.) Perform DETAILED CAL.

B.) Perform AUTO CT# ADJUST.

C.) Repeat this procedure to verify 48cm Phantom Image Performance.

6.5 20cm QA Phantom Image Series Image Performance Verification

6.5.1 Acquiring the 20cm QA Phantom Image Series

1.) Mount the Phantom Holder on the head-end of the table.

2.) Mount the 20cm QA Phantom on the Phantom Holder.

3.) Align, level, & center the 20cm QA Phantom.

- Align the etched line (QA#1 position) on phantom using the internal laser lights.
- Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
- Center phantom using the CENTER PHANTOM procedure in the left head SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.

4.) Set up the system to scan the QA#1, QA#2, and QA#3 positions on the 20cm QA Phantom.

Manual Scan Protocol Setup

Refer to Table 7 and set-up an Axial scan with the parameters shown

Table 7: LightSpeed Plus 20cm QA Phantom Image Series Scan Parameters

Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
QA#1	Axial Full 1.0 sec.	I5.00	S5.00	4	10.0 2i	0.00	S0.0	Small	120	260	2.0 sec.	25.0	Std
Recon 2: Q	Recon #2 of Series #1–4 images											15.0	Bone
QA#2	Axial Full 1.0 sec.	S40.00	S50.00	8	10.0 2i	0.00	S0.0	Small	120	260	4.0 sec.	25.0	Std
QA#3	Axial Full 1.0 sec.	S55.00	S65.00	8	10.0 2i	0.00	S0.0	Small	120	260	4.0 sec.	25.0	Std

Auto Scan Protocol Setup

a.) On the Exam Rx desktop, select NEW PATIENT.

b.) Type the following entries in the listed `Patient Information` following fields:

* `Patient ID:`**`Service`**
* `Name:` **`20cm QA#3 Phantom Image Series`**

c.) From the Protocol display, click on the infant box.

d.) On the Infant display window, click on the area below the infant's right foot to display the `Miscellaneous` menu.

e.) Click on the 20:12 IMAGE SERIES QA button.

f.) When the scan prescription appears on the left monitor, select the 1st series.

5.) Set internal Landmark.

6.) Acquire the QA#1, QA#2 and QA#3 positions of the 20cm QA Phantom.

6.5.2 High Contrast Spatial Resolution

6.5.2.1 Image Performance Verification

1.) Select the first 20cm QA Phantom image series (QA#1) from the exam acquired in the previous section.

a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.

b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.

c.) Select the exam and the first series acquired in the previous section.

d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.

2.) Build a 17 x 17 pixel reference ROI Box using the Image Works Viewer tools.

a.) Click on the grid button to place a grid on the first image.

b.) Click on the MEASURE button and select the box ROI icon.

c.) Adjust the size of the ROI box to be a square of approximately 8 mm x 8 mm (64 mm^2) box.

If required, magnify the image to adjust to proper dimensions. See Figure 6 for additional ROI size and placement information.

3.) Collect Mean and Standard deviation values for three reference ROI Box positions on the four 20cm QA#1 Phantom images.

If required, magnify the image to adjust to proper dimensions. See Figure 6 for additional ROI placement information.

a.) Position the reference ROI Box built in step 2 directly over the 1.6 mm Line Pair Pattern on the QA#1 phantom image. Fine adjust size of Box #1 ROI to ensure that the box does not exceed the line pair boundaries. See Figure 6.

b.) Type **`prop a`** in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.

c.) Click on the MEASURE button and select the box ROI icon to display Box # 2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box # 1.

d.) Reposition Box # 2 over the Plexiglas portion of the QA#1 phantom image (Box # 2 position in Figure 6).

e.) Type **`prop a`** in the Image Works Accelerator Line followed by <Return> to propagate Box # 2 ROI (size and position) on the remaining images.

f.) Click on the MEASURE button and select the box ROI icon to display Box # 3. The system places an ROI box labeled # 3 at the center of the image with the exact same dimensions as Boxes # 1 and 2.

g.) Reposition Box # 3 over the water portion of the QA#1 phantom image (Box # 3 position in Figure 6).

h.) Record the Mean and Standard Deviation values of the four images in the series for each of the three box positions in Table 8.

Figure 6: 20cm QA#1 Phantom High Contrast Spatial Resolution Testing - Building & Placing Reference ROI Boxes

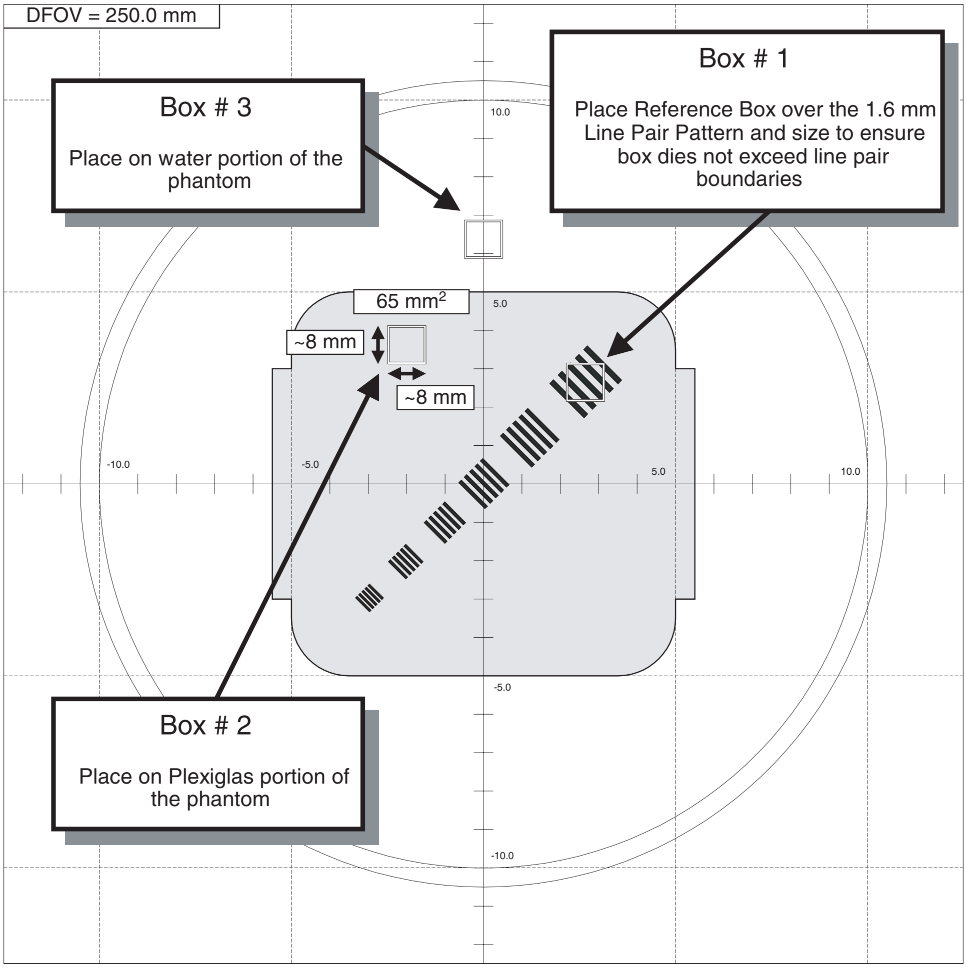


Table 8: 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance Worksheet #1

Image	Box 1		Box 2		Box 3		Contrast Scale (Box 2 Means – Box 3 Means)	Std Dev Average (Box 2 Std Dev – Box 3 Std Dev / 2)
	Means	Std Dev	Means	Std Dev	Means	Std Dev		
1								
2								
3								
4								

Table 9: 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance Worksheet #2

Image	MTF	MTF 4-slice average	Contrast Scale	Comments
1		n/a		
2		n/a		
3		n/a		
4		n/a		
Specifications	n/a	0.65 to 1.0	110.0 to 130.0	n/a

4.) Calculate the Contrast Scale value and four-image average MTF values for each image in the series, record values in Table 9, and compare to specifications.

a.) Calculate and record the Contrast Scale value (Means value of Box 2 - Means value of Box 3) for each image and record in Table 8 and Table 9.

b.) Calculate and record the MTF value for each image in Table 9 using the following formula. Record Standard Deviation Average (Std Dev AVE) values in both Table 8 and Table 9.

Std Dev AVE = $(\text{Std Dev}_{\text{Box 2}} - \text{Std Dev}_{\text{Box 3}}) + 2$

Modulation = $\sqrt{\text{StdDev}^2_{\text{Box1}} - \text{StdDev}^2_{\text{AVE}}}$

MTF = (2.2 x Modulation) + Contrast Scale

c.) Calculate the average MTF value for each image and record in Table 9.

d.) Verify Contrast value for each image and the four-image MTF value meets specifications listed in Table 9.

e.) Record the 20cm QA Phantom High Contrast Spatial Resolution values (Contrast Scale for each image and four-image MTF value) in the HHS Record Tables.

5.) View each image of the Recon 2 Series, record all the visible Line Pair Patterns in Table 10, and verify each image meets specifications.

a.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.

b.) Select the exam and the second series (Recon 2: Q) acquired in the previous section.

c.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.

d.) While viewing each image, indicate in Table 10 all the Line Pair Pattern groups that can be visually distinguished. See Figure 7.

e.) Verify Line Pair Pattern visual check meets specifications and record in the HHS Record Tables.

Table 10: 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance Worksheet #3

Image	Line Patterns Visible	Comments
1		
2		
3		
4		
Specifications	B, C, D, E, F	n/a

6.5.2.2 Failure Recovery

Specifications

Each image of the series must pass 20cm QA#1 High Contrast Spatial Resolution parameter (Contrast Scale, Four-Image MTF Average and Visible Line Pair (for the first and second series scan parameters) specifications:

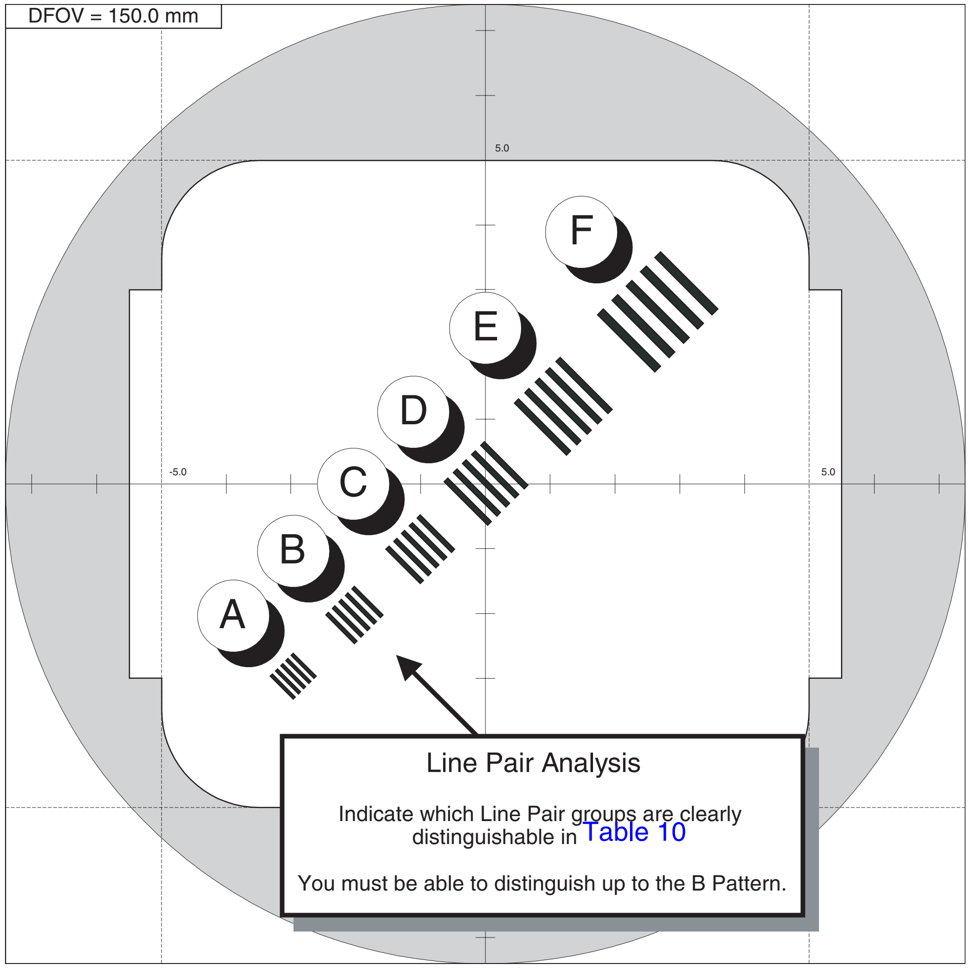
- Contrast Scale: 110.0 to 130.0
- MTF Average: 0.65 to 1.0
- Visible Lines: B, C, D, E, F

Recommended Recovery

A.) Check Phantom Alignment (leveling is critical) and repeat this scanning and High Contrast Spatial Resolution Performance Test.

B.) Perform Alignment Procedures (POR Alignment, BOW Alignment, CBF/SAG Alignment, ISO Alignment, and Hot ISO Alignment), perform Full Calibration (Detailed Calibration and Auto CT# Adjust, and repeat this scanning and High Contrast Spatial Resolution Performance Test.

Figure 7: 20cm QA#1 Phantom High Contrast Spatial Resolution (Visible Line Verification)



6.5.3 Low Contrast Detectability

6.5.3.1 Image Performance Verification

1.) Select the third 20cm QA Phantom image series (QA#2 Holes) from the exam acquired in the previous section.
   a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
   b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
   c.) Select the exam and the third series acquired in the previous section.
   d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.
2.) Build a 10 mm x 100 mm reference ROI Box using the Image Works Viewer tools.
   a.) Click on the grid button to place a grid on the first image.
   b.) Click on the MEASURE button and select the box ROI icon.
   c.) Adjust the size of the ROI box to be a rectangle of approximately 10 mm x 50 mm (500 mm2) box.

   If required, magnify the image to adjust to proper dimensions. See Figure 8 for additional ROI size and placement information.
3.) Collect Mean values for two reference ROI Box positions on four (1st, 3rd, 5th, and 7th) of the eight 20cm QA#2 Phantom images.

   If required, magnify the image to adjust to proper dimensions. See Figure 8 for additional ROI placement information.
   a.) Adjust the Window setting of the image to 20 by simultaneously holding the center mouse button down while dragging the cursor to the left. Adjust level for a viewable image as shown in Figure 8.
   b.) Position the reference ROI Box built in step 2 on the Plexiglas portion of the QA#2 phantom image without touching the water portion or the hole pattern. See Figure 8.
   c.) Type **`prop a`** in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
   d.) Click on the MEASURE button and select the box ROI icon to display Box # 2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box # 1.
   e.) Reposition Box # 2 over the water portion of the QA#1 phantom image and directly above Box # 1 (Box # 2 position in Figure 8).
   f.) Type **`prop a`** in the Image Works Accelerator Line followed by <Return> to propagate Box # 2 ROI (size and position) on the remaining images.
   g.) Record the Mean values of the 1st, 3rd, 5th, and 7th images in the series for each of the two box positions in Table 8.

Figure 8: 20cm QA#2 Phantom Low Contrast Detectability - Building and Placing Reference ROI Boxes

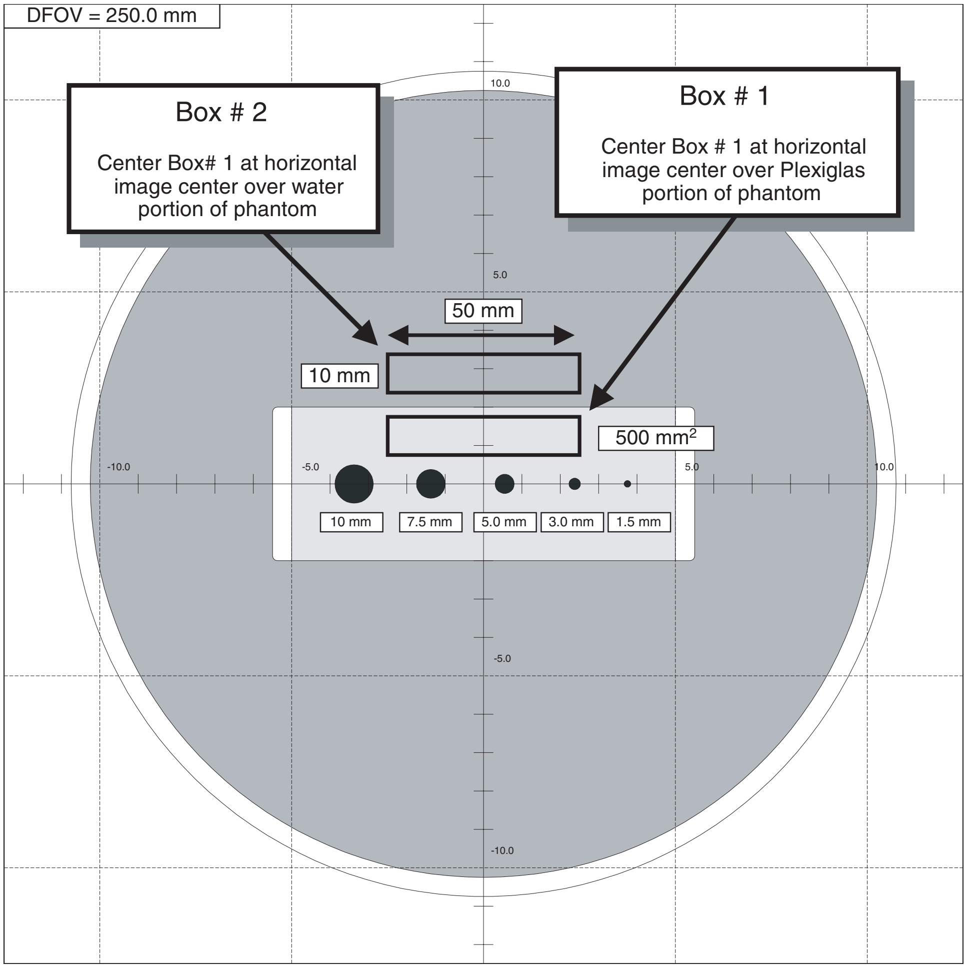


Table 11: 20cm QA#2 Phantom Low Contrast Detectability Image Performance Worksheet #1

Image	Visible Holes Viewable at Window 20	Box 1 Means (Plexiglas)	Box 2 Means (Water)	Contrast Factor (Box 1 Means – Box 2 Means)	Comments
1					
3					
5					
7					
Specifications	See Table 12	n/a	n/a	2.0 to 12.0	n/a

4.) View the 1st, 3rd, 5th, and 7th images of the QA#2 Holes Series, record the number of visible holes in Table 11, and verify each image meets specifications.

a.) While viewing the 1st, 3rd, 5th, and 7th images, indicate in Table 11 the number of holes that can be visually distinguished. See Figure 8.

b.) Verify Visible Hole visual check meets specifications listed in Table 12 for the calculated Contrast Factor and record in the HHS Record Tables.

Table 12: 20cm QA#2 Phantom Visible Hole Specifications

Contrast Factor Range (Box 1 Means – Box 2 Means)	Visible Number of Holes		Smallest Visible Hole Size
	Lower Limit*	Upper Limit*	
2.00 to 3.99	2	5	7.5mm
4.00 to 7.99	3	5	5.0mm
8.00 to 12.00	4	5	3.0mm
*Required number of visible holes depends on the contrast factor.			

6.5.3.2 Failure Recovery

Specifications

At least two out of the four images of the series must pass the 20cm QA#2 Low Contrast Detectability parameter (Visible Hole Size for a calculated Contrast Factor) for the third series scan parameters specifications.

Recommended Recovery

A.) Check Phantom Alignment (leveling is critical) and repeat this scanning and High Contrast Spatial Resolution Performance Test.

B.) Perform Alignment Procedures (POR Alignment, BOW Alignment, CBF/SAG Alignment, ISO Alignment, and Hot ISO Alignment), perform Full Calibration (Detailed Calibration and Auto CT# Adjust, and repeat this scanning and Low Contrast Detectability Performance Test.

6.5.4 QA#3 Phantom Brightness Uniformity and CT#

6.5.4.1 Performance Verification

1.) Select the fourth 20cm QA#3 Phantom exam acquired in the previous section.
   - a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
   - b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
   - c.) Select the exam and fourth (QA#3) series acquired in the previous section.
   - d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.

2.) Build a 31 x 31 pixel reference ROI Box using the Image Works Viewer tools.
   - a.) Click on the grid button to place a grid on the first image.
   - b.) Click on the MEASURE button and select the box ROI icon.
   - c.) Adjust the size of the ROI box to be a square 15 mm x 15 mm (225 mm2) box. Tolerance: 15 mm +/- 1 mm (196 mm2 to 256 mm2).

     If required, magnify the image to adjust to proper dimensions. See Figure 9 for additional ROI size and placement information.

3.) Collect Mean values for five reference ROI Box positions on the eight 20cm QA#3 Phantom images.

   If required, magnify the image to adjust to proper dimensions. See Figure 9 for additional ROI placement information.
   - a.) Position the reference ROI Box built in step 2 directly over the center of the image using the grid cross-hairs as a guide.
   - b.) Type `prop a` in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
   - c.) Click on the MEASURE button and select the box ROI icon to display Box # 2. The system places an ROI box labeled # 2 at the center of the image with the exact same dimensions as Box # 1.

d.) Reposition Box # 2 to the left center portion on the first image (Box # 2 position in Figure 5).

e.) Type **prop a** in the Image Works Accelerator Line followed by <Return> to propagate Box # 2 ROI (size and position) on the remaining images.

f.) Repeat steps 3.c. through 3.e. for each of the remaining ROI box positions shown in Figure 6 (Box # 3 through 5).

g.) Record the Mean values of the eight images in the series for each of the five box positions in Table 13.

Each image can only display text for the mean, standard deviation, and box area for three images at a time. To view the data for a particular box, select the box on the image and the system displays the data for the box number selected.

Figure 9: 20cm QA#3 Phantom Brightness Uniformity & CT# Measurement - Building And Placing Reference ROI Boxes

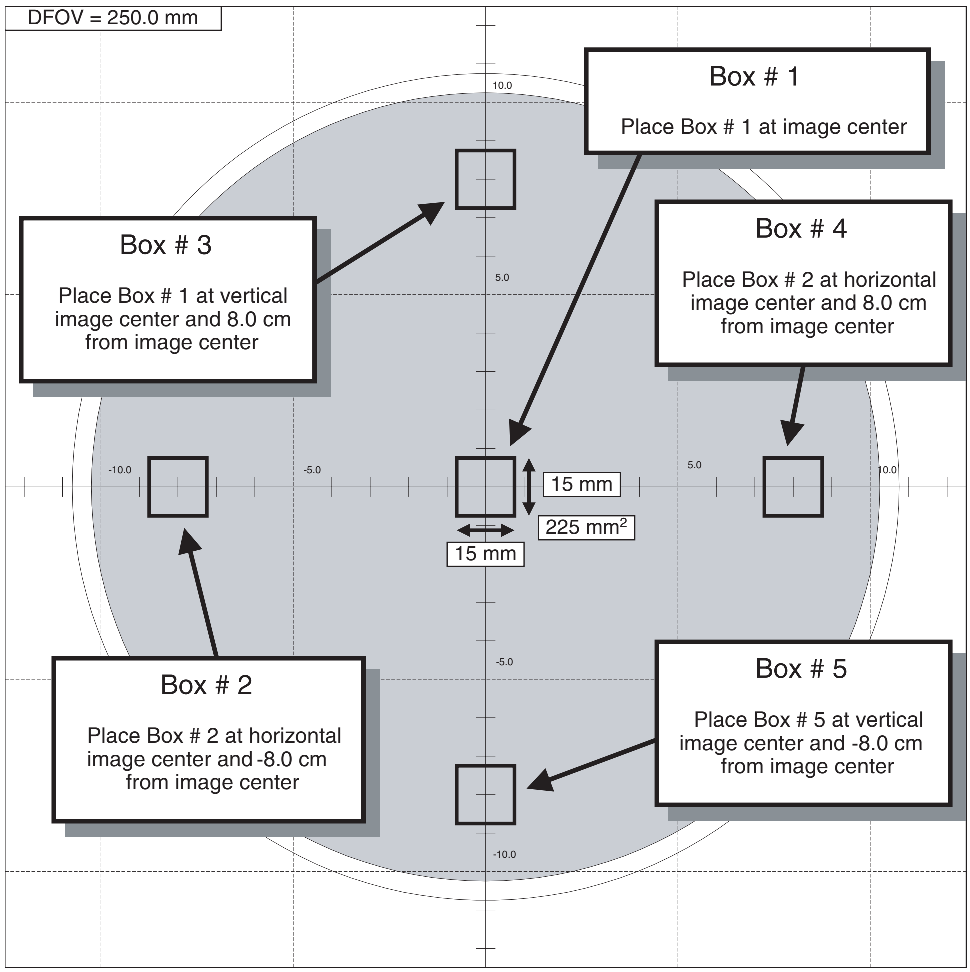


Table 13: 20cm QA#3 Phantom CT# Brightness Uniformity & CT# Image Performance Worksheet

Image	Box 1 (Center)	Box 2 (Left Center)		Box 3 (Top Center)		Box 4 (Right Center)		Box 5 (Bottom Center)		Outside Boxes Avg	
	Means (Xc)	Means	Std dev	Means	Std dev	Means	Std dev	Means	Std dev	Means (AvXo)	Std dev (AvSDo)
1			n/a		n/a		n/a		n/a		n/a
2			n/a		n/a		n/a		n/a		n/a
3			n/a		n/a		n/a		n/a		n/a
4			n/a		n/a		n/a		n/a		n/a
5			n/a		n/a		n/a		n/a		n/a
6			n/a		n/a		n/a		n/a		n/a
7			n/a		n/a		n/a		n/a		n/a
8			n/a		n/a		n/a		n/a		n/a

Box Size =196 mm2 to 256 mm2

15 mm (+/- 1 mm) x 15 mm (+/- 1 mm)

31 (+/- 2 pixels) x 31 (+/- 2 pixels)

Box Positions:Box 1 = 0 mm x 0 mm

Box 2 = 0 mm x -80 mm

Box 3 = 80 mm x 0 mm

Box 4 = 0 mm x 80 mm

Box 5 = -80 mm x 0 mm

Table 14: 48cm Phantom CT# Brightness Uniformity & Noise Row Performance Worksheet

Row	Images	Center Box Means (AvXc)	Outer Boxes Means Averages (AvXo)	Outer Boxes Means Averages minus Center Box Means (AvXo - AvXc)	Outer Boxes Standard Deviation Averages (AvSDo)	Comments
2A1A	1, 3, 5, 7				n/a	
1B2B	2, 4, 6, 8				n/a	
Specifications		0.0 ± 3.0	n/a	<± 3.0	n/a	

4.) Calculate the Brightness Uniformity and CT# values for each image in the series, record values in Table 14, and compare to specifications.

a.) Calculate and record the average means values (AvXo) for the four outside Boxes (Boxes 2 through 5) for each of the images and record in Table 13.

b.) Calculate and record the average center box (Box 1) means values (AvXc) for each row (2A1A: Images 1, 3, 5, and 7; 1B2B: Images 2, 4, 6, 8) in Table 14.

c.) Calculate and record the average outside boxes (Box 2 through 5) means values (AvXo) for each row (2A1A: Images 1, 3, 5, and 7; 1B2B: Images 2, 4, 6, 8) in Table 14.

d.) Calculate the Brightness Uniformity (AvXo - AvXc) value for each row and record in Table 14.

e.) Verify Brightness Uniformity (AvXo - AvXc) value and the average CT# value (AvXc) for each row meets specifications listed in Table 14.

f.) Record the 20cm QA#3 Phantom Brightness Uniformity (AvXo - AvXc) value and average CT# value (AvXc) for each row in the HHS Record Tables.

6.5.4.2 Failure Recovery

Specifications

Each Row (2A1A, 1B2B) of the series must pass 20cm QA#3 Phantom Brightness Uniformity and average CT# specifications:

- AvXo - AvXc: < +/- 3.0
- AvXc: < +/- 3.0

Recommended Recovery

A.) Perform DETAILED CAL.

B.) Perform AUTO CT# ADJUST.

C.) Repeat this procedure to verify 20cm QA#3 Phantom Image Performance.

6.5.5 QA#3 Phantom Noise

6.5.5.1 Performance Verification

1.) Select the fourth 20cm QA#3 Phantom exam acquired in the previous section.
   a.) From the Global Control Palette, click on the IMAGE WORKS Desktop.
   b.) From the Image Works Desktop, select the IMAGE WORKS BROWSER window.
   c.) Select the exam and fourth (QA#3) series acquired in the previous section.
   d.) Select the VIEWER button on the Image Works Browser window. Set up the viewer window for four-image viewing.

2.) Build a 51 x 51 pixel reference ROI Box using the Image Works Viewer tools.
   a.) Click on the grid button to place a grid on the first image.
   b.) Click on the MEASURE button and select the box ROI icon.
   c.) Adjust the size of the ROI box to be a square 25 mm x 25 mm (625 mm2) box. Tolerance: 25 mm +/- 1 mm (576 mm2 to 676 mm2).

   If required, magnify the image to adjust to proper dimensions. See Figure 10 for additional ROI size and placement information.

3.) Collect Standard Deviation values for the single reference ROI Box position on the eight 20cm QA#3 Phantom images.

   If required, magnify the image to adjust to proper dimensions. See Figure 10 for additional ROI placement information.

   a.) Position the reference ROI Box built in step 2 directly over the center of the image using the grid cross-hairs as a guide.
   b.) Type `prop a` in the Image Works Accelerator Line followed by <Return> to propagate Box # 1 ROI (size and position) on the remaining images in the series.
   c.) Record the Standard Deviation values of the eight images in the series for centered box position in Table 15.

4.) Calculate the average Noise values for each image in the series, record values in Table 15, and compare to specifications.
   a.) Calculate and record the average Noise values (AvSDc) for the inside Boxes for each of the two rows (2A1A and 1B2B) and record in Table 15.
   b.) Verify Noise (AvSDc) values for each row meets specifications listed in Table 15.
   c.) Record the 20cm QA#3 Phantom Noise (AvSDc) for each row in the HHS Record Tables.

Figure 10: 20cm QA#3 Noise Measurement - Building And Placing Reference ROI Box

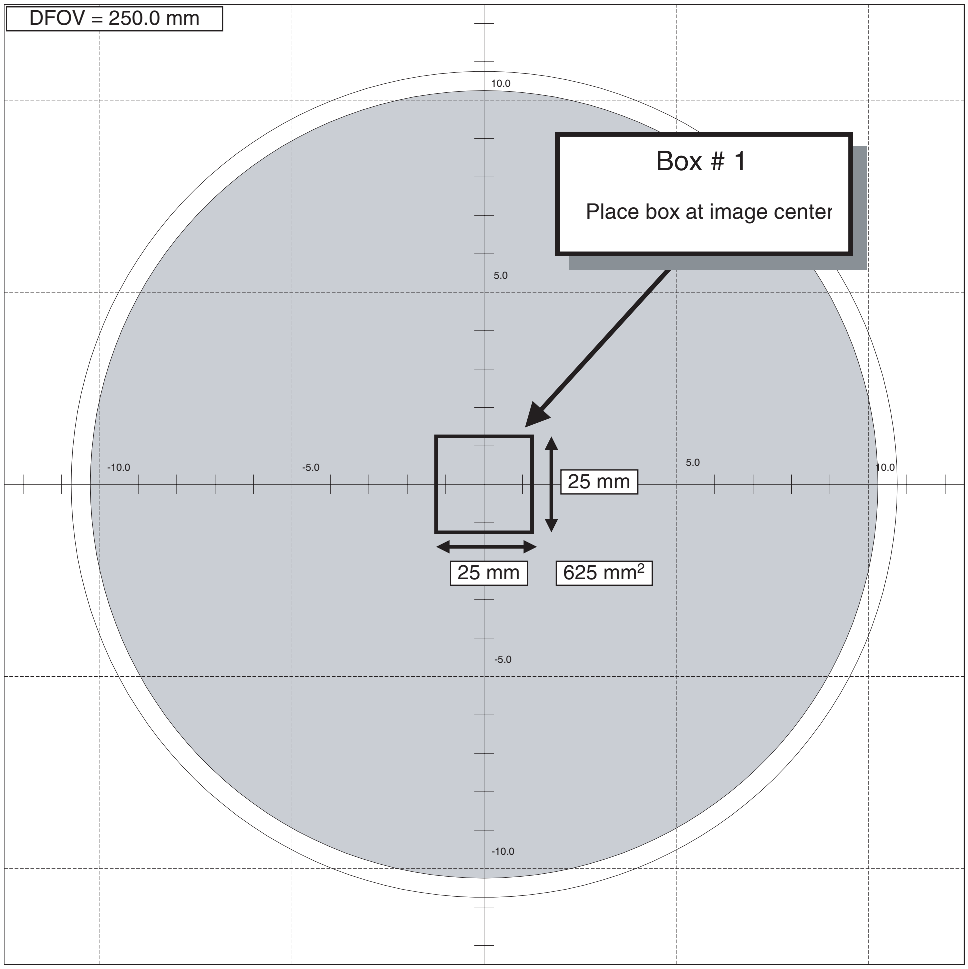


Table 15: 20cm QA#3 Phantom Noise Performance Worksheet

Image	Row	Box 1 Std Dev (SDc)	Average Std Dev (AvSDc)	AvSDc Specifications		Comments
				Systems with less than 5000 scans	Systems with more than 5000 scans	
1	2A1A			3.2 ± 0.3	3.2 ± 0.4	
3						
5						
7						
2	1B2B					
4						
6						
8						

Box Size =576 mm2 to 676 mm2

25 mm (+/- 1 mm) x 25 mm (+/- 1 mm)

51 (+/- 2 pixels) x 51 (+/- 2 pixels)

Box Position:Box 1 = 0 mm x 0 mm

6.5.5.2 Failure Recovery

Specifications

Each Row (2A1A, 1B2B) of the series must pass 20cm QA#3 Phantom Noise specifications shown in Table 15.

Recommended Recovery

A.) Perform DETAILED CAL.

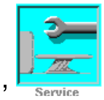
B.) Perform AUTO CT# ADJUST.

C.) Repeat this procedure to verify 20cm QA#3 Phantom Image Performance.

7 Save System State

Use the following commands to create the System State MOD.

1.) Load a MOD into the mod drive on the front of the console.

2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon, 

.

3.) Click on the PM icon. 

4.) Select SYSTEM STATE to open the System State Save/Restore menu.

5.) Select ALL

6.) Select SAVE

7.) When the save operation completes, select FILE and QUIT from the pull down menu.

8.) Remove the MOD from the drive